

Impact Statement Card

Values-led impact assessment for early detection of malicious/harmful LLM prompts

Project	Detecting harmful/malicious user intent via internal LLM representations
Team	Komadugu Team
Method	Linear probe on frozen Gemma-2-2B layer-14 embeddings, trained on Civil Comments
Card version	1.0

1. Project Summary

Large language models can sometimes respond to harmful or malicious prompts, and today's guardrails often act during or after generation, making them costly and hard to interpret. This project probes a model's **internal representations** to flag likely-harmful intent *before* a response is generated, using a lightweight, interpretable classifier (StandardScaler + Logistic Regression) on frozen Gemma-2-2B hidden states, as a faster and more transparent complement to output-side moderation.

2. Positive Impact — Who Benefits, and How

Stakeholder	Potential Benefit
End users of AI systems	Fewer harmful responses reach them; faster, cheaper safety checks reduce latency/cost that might otherwise limit safety coverage.
Organizations deploying LLMs	A low-cost, interpretable pre-generation signal that can complement existing moderation stacks and reduce legal/reputational exposure.
AI developers & safety researchers	Evidence about where/how safety-relevant concepts are encoded inside an LLM (interpretability value beyond this one classifier).
Vulnerable/at-risk online communities	Reduced exposure to hate speech, harassment, and dangerous instructions if the approach is integrated responsibly.

3. Potential Risks & Negative Impacts

- **False positives** — harmless prompts (e.g., medical, legal, or creative-writing questions) misclassified as toxic, degrading legitimate user experience or blocking access.
- **False negatives** — genuinely harmful prompts pass undetected, giving a false sense of safety if the probe is over-trusted.
- **Disparate impact** — the underlying Civil Comments data has documented identity-term bias; a probe trained on it can inherit over-flagging of comments mentioning certain identities, religions, or dialects.
- **Privacy** — prompts are necessarily processed by the backbone model to produce embeddings; any logging of prompts or embeddings for debugging must be minimized and secured.
- **Automation bias** — downstream systems may over-rely on an automated flag instead of human judgment for high-stakes or ambiguous cases.

- **Limited explainability** — a linear probe on a hidden layer is more interpretable than the full LLM, but an end user told "flagged as harmful" still cannot easily see why.
- **Dual-use / misuse** — probing techniques that reveal how safety concepts are encoded could, in principle, be adapted to search for ways to evade detection.

4. Values Guiding the Solution

Value	What it means for this project
Safety	Prevent harmful AI interactions before generation, without introducing new harms.
Fairness	Avoid disproportionately flagging particular users, languages, or communities.
Transparency	Keep the detector interpretable (linear probe on a documented layer) rather than an opaque black box.
Privacy	Avoid unnecessary collection or retention of user prompt data.
Accountability	Make the system's decisions auditable and its limitations documented (this card, the Data Card).
Reliability	Consistent performance across varied, realistic user inputs, verified through held-out evaluation.

5. Mitigations

- Evaluate with both false-positive and false-negative rates, not accuracy alone (see project README/evaluation results).
- Report class-imbalance handling (class-weighted logistic regression) and validate on a held-out split.
- Document known dataset bias (Data Card §7) so downstream integrators can plan additional fairness auditing before deployment.
- Keep the classifier head simple and inspectable; publish the probed layer and pooling method.
- Recommend human oversight for borderline-probability or high-stakes classifications rather than fully automated blocking.
- No prompt text or embeddings are persisted by the reference Streamlit app beyond the active session unless a user explicitly exports results.

6. Success Criteria

- Accurate detection of harmful prompts with low false-positive and false-negative rates (not accuracy alone).
- Comparable performance across the identity/topic subgroups represented in the data.
- Lower computational cost than full generate-then-filter moderation pipelines.
- Actionable interpretability: a clear, documented answer to "which layer/representation drives this decision."
- A design that could plausibly be integrated into a real application alongside human oversight, without blocking legitimate use.

7. Residual Risk Statement

This project is a research prototype trained on a single English-language dataset with documented bias limitations, evaluated on ~5,000 held-out comments. It has not been audited for production fairness or adversarial robustness, and should not be used as a sole or final safety gate in any deployed system.

Derived from the Komadugu Team's Values-Led Problem Statement (Challenge 1) and the project's data/training pipeline. See [data_card.pdf](#) and [stakeholder_engagement.pdf](#) for supporting detail.